Capability Consolidation and Deployment-Aware Post-Training

Deep Dive into Kimi K3 — Chapter 23

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Foundation Books Project

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Goal: route each student prefix to its cell's teacher, keep every cell and effort level visible, train through the quantized forward with parity-gated updates, then freeze the target and train a draft for acceptance — each claim in its evidence class.

Roadmap

Routed on-policy distillation

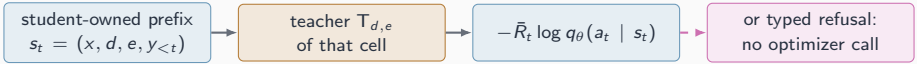
Retention and effort control

Precision inside learning

Acceptance-optimized drafts

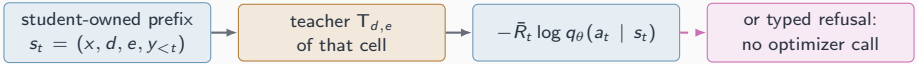
Routed on-policy distillation

Student prefixes, routed teachers, detached coefficients



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- K3 reports nine domain–effort specialists consolidated by MOPD. [REPORTED: K3 Sec. 4.1.2–4.1.3] Binary fixture: $q(\mathrm{below})$ 0.55 $\rightarrow$ 0.557252.

Retention and effort control

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  [OBSERVED: small-model studies]

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- Interference is a matched contrast; effort labels are requests, resource vectors are behavior.

Precision inside learning

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- Strict parity: equal ordered fingerprints, then replayed decisions, then tolerance. MX learning study: 79/128 (QAT) vs. 80/128 (float32-then-MX) — learning, no QAT advantage. [OBSERVED: book MX study]

Acceptance-optimized drafts

Freeze the target, train for acceptance

- $\alpha(p, q) = \sum_v \min\{p(v), q(v)\} = 1 - \mathrm{TV}(p, q)$; residual correction restores the target law at one prefix. Fixture: $\alpha = 0.9$, $L_{\mathrm{LK}} = 0.10536$.

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A correct calculation is not a passing gate. Next: Chapter 24's Unified Agentic Environments and Task Synthesis.